

Lecture 15 - Class Activity ANCOVA

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Lecture 15: Analysis of Covariance (ANCOVA)

What is ANCOVA?

ANCOVA (Analysis of Covariance) combines regression and ANOVA to: - Compare group means while adjusting for a continuous covariate - Increase statistical power by reducing residual error - Control for confounding variables

When to Use ANCOVA

Use ANCOVA when you have: - **Response variable**: Continuous - **Predictor variable**: Categorical (factor/groups) - **Covariate**: Continuous variable that affects the response

Key Assumptions of ANCOVA

1. **Independence** of observations
2. **Normality** of residuals
3. **Homogeneity of variances** across groups
4. **Linearity** between response and covariate within each group
5. **Homogeneity of slopes** (most critical!) - regression slopes must be equal across all groups

! Critical First Step

Always test for **homogeneity of slopes** before proceeding with ANCOVA. If slopes differ significantly between groups, standard ANCOVA is inappropriate.

Part 1: Cricket Chirping Analysis

Data Overview

We want to compare chirping rate of two cricket species: - *Oecanthus exclamationis* - *Oecanthus niveus*

But we measured rates at different temperatures, and there's a relationship between pulse rate and temperature. ANCOVA lets us adjust for temperature effect to get a more powerful test!

```
# Create simulated cricket data based on lecture example
set.seed(456)
n <- 40
species <- rep(c("0. exclamationis", "0. niveus"), each = n/2)
temp <- c(rnorm(n/2, mean = 22, sd = 2), rnorm(n/2, mean = 24, sd = 2))
chirp_rate <- 40 + 2.5 * (temp - 23) + ifelse(species == "0. exclamationis", 10, 0) +
  rnorm(n, sd = 3)
cricket_data <- data.frame(species = species, temp = temp, chirp_rate = chirp_rate)

# View data structure

head(cricket_data)
```

```

      species      temp chirp_rate
1 0. exclamationis 19.31296   40.69557
2 0. exclamationis 23.24355   51.78799
3 0. exclamationis 23.60175   50.75553
4 0. exclamationis 19.22222   40.80589
5 0. exclamationis 20.57129   50.16484
6 0. exclamationis 21.35188   46.24225

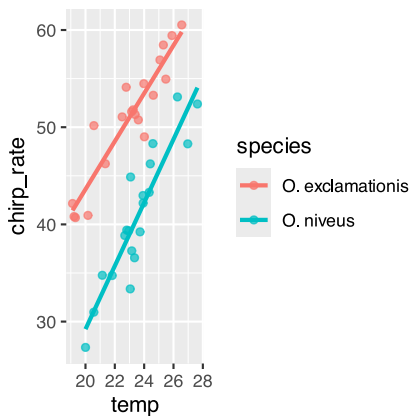
```

```

# Plot with regression lines by species
ggplot(cricket_data, aes(x = temp, y = chirp_rate, color = species)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE)

```

`geom\_smooth()` using formula = 'y ~ x'



Step 1: Test Homogeneity of Slopes

This is the most critical assumption! We test if the regression slopes are equal across all groups.

```

# Test for homogeneity of slopes by including interaction term
cricket_slopes_model <- lm(chirp_rate ~ temp * species, data = cricket_data)
Anova(cricket_slopes_model, type = 3)

```

Anova Table (Type III tests)

Response: chirp_rate

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	6.32	1	0.9393	0.338915
temp	620.48	1	92.1572	0.0000000000001828 ***
species	69.76	1	10.3617	0.002724 **
temp:species	26.08	1	3.8734	0.056796 .
Residuals	242.38	36		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Interpretation: If $p > 0.05$, slopes are homogeneous and we can proceed with ANCOVA. If $p < 0.05$, slopes differ and standard ANCOVA is inappropriate.

Step 2: Fit ANCOVA Model

Since slopes are homogeneous ($p > 0.05$), we can fit the ANCOVA model without the interaction term.

```
# Fit ANCOVA model (without interaction)
cricket_ancova <- lm(chirp_rate ~ temp + species, data = cricket_data)

# Get model summary
summary(cricket_ancova)
```

```
Call:
lm(formula = chirp_rate ~ temp + species, data = cricket_data)

Residuals:
    Min       1Q   Median       3Q      Max
-6.0065 -1.9653  0.1923  0.7886  5.9192

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   -13.2012     4.7423  -2.784   0.00842 **
temp             2.7926     0.2048  13.634 0.000000000000000530 ***
species0. niveus -11.8005     0.8593 -13.733 0.000000000000000424 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.694 on 37 degrees of freedom
Multiple R-squared:  0.8994,    Adjusted R-squared:  0.894
F-statistic: 165.4 on 2 and 37 DF,  p-value: < 0.00000000000000022
```

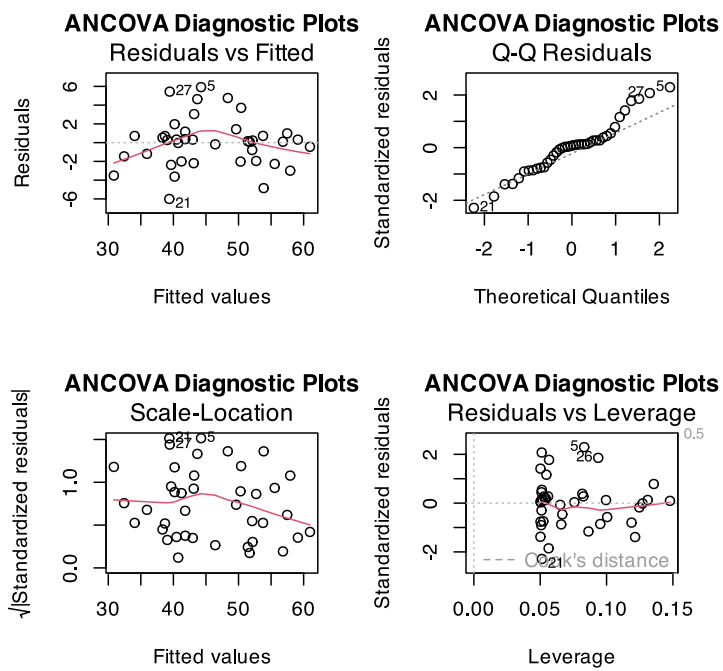
```
# View ANOVA table
Anova(cricket_ancova)
```

```
Anova Table (Type II tests)

Response: chirp_rate
      Sum Sq Df F value    Pr(>F)
temp    1348.81  1  185.90 0.0000000000000005296 ***
species  1368.34  1  188.59 0.0000000000000004236 ***
Residuals  268.46 37
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Step 3: Check Model Assumptions

```
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(cricket_ancova, main = "ANCOVA Diagnostic Plots")
```



```
par(mfrow = c(1, 1))
```

Step 4: Calculate Adjusted Means

ANCOVA compares adjusted means - what each group's mean would be at the overall mean of the covariate.

```
# Calculate adjusted means using emmeans
cricket_adjusted_means <- emmeans(cricket_ancova, "species")

# Convert to dataframe for plotting
cricket_adj_means_df <- as.data.frame(cricket_adjusted_means)
cricket_adj_means_df
```

species	emmean	SE	df	lower.CL	upper.CL
0. exclamationis	51.70513	0.6049702	37	50.47934	52.93091
0. niveus	39.90462	0.6049702	37	38.67883	41.13040

Confidence level used: 0.95

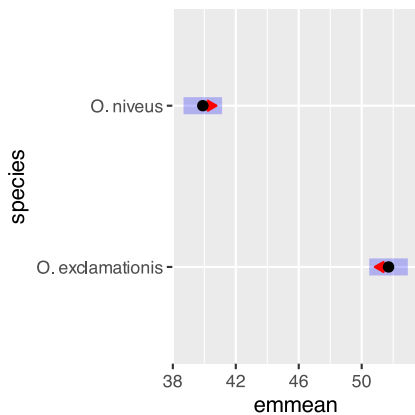
Step 5: Pairwise Comparisons

```
# Pairwise comparisons of adjusted means
pairs(cricket_adjusted_means, adjust = "sidak")
```

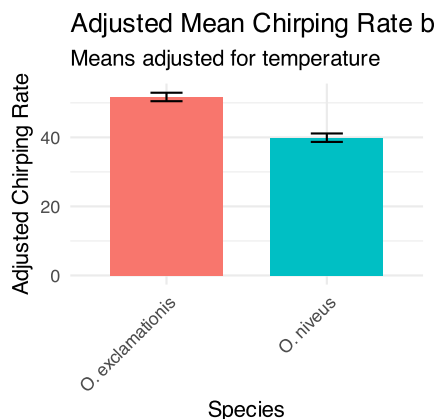
contrast	estimate	SE	df	t.ratio	p.value
0. exclamationis - 0. niveus	11.8	0.859	37	13.733	<.0001

Step 6: Visualize Results

```
# Plot adjusted means with confidence intervals
plot(cricket_adjusted_means, comparisons = TRUE)
```



```
# Bar chart of adjusted means
ggplot(cricket_adj_means_df, aes(x = species, y = emmean, fill = species)) +
  geom_bar(stat = "identity", width = 0.7) +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.2) +
  labs(title = "Adjusted Mean Chirping Rate by Species",
       subtitle = "Means adjusted for temperature",
       x = "Species",
       y = "Adjusted Chirping Rate") +
  theme_minimal() +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1))
```



Part 2: Partridge Longevity Analysis

Data Overview

We'll analyze the effect of mating strategy on male fruitfly longevity, using thorax length as a covariate.

```
# Load the partridge dataset
partridge <- read.csv("data/partridge.csv")

# Create better treatment names
partridge$treatment <- factor(partridge$TREATMEN,
                              levels = 1:5,
                              labels = c("No females",
                                           "One virgin female daily",
                                           "Eight virgin females daily",
                                           "One inseminated female daily",
```

```
"Eight inseminated females daily"))
```

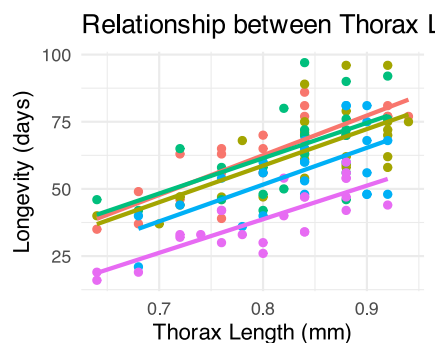
```
# View data structure
head(partridge)
```

	PARTNERS	TYPE	TREATMEN	LONGEV	LLONGEV	THORAX	RESID1	PREDICT1	RESID2
1	8	0	1	35	1.544068	0.64	-5.868456	40.86846	-0.04743024
2	8	0	1	37	1.568202	0.68	-9.301196	46.30120	-0.07105067
3	8	0	1	49	1.690196	0.68	2.698804	46.30120	0.05094369
4	8	0	1	46	1.662758	0.72	-5.733936	51.73394	-0.02424867
5	8	0	1	63	1.799341	0.72	11.266064	51.73394	0.11233405
6	8	0	1	39	1.591065	0.76	-18.166676	57.16668	-0.14369601

	PREDICT2	treatment
1	1.591498	No females
2	1.639252	No females
3	1.639252	No females
4	1.687007	No females
5	1.687007	No females
6	1.734761	No females

```
# Visualize the relationship between thorax length and longevity by treatment
ggplot(partridge, aes(x = THORAX, y = LONGEV, color = treatment)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Relationship between Thorax Length and Longevity",
       x = "Thorax Length (mm)",
       y = "Longevity (days)",
       color = "Treatment") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



nale daily — Eight virgin females daily — One ins

Step 1: Test Homogeneity of Slopes

```
# Test for homogeneity of slopes
homo_slopes_model <- lm(LONGEV ~ THORAX * treatment, data = partridge)
Anova(homo_slopes_model, type = 3)
```

Anova Table (Type III tests)

Response: LONGEV

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	755.6	1	6.6320	0.01128 *
THORAX	3486.3	1	30.5999	2.017e-07 ***
treatment	36.9	4	0.0810	0.98805
THORAX:treatment	42.5	4	0.0933	0.98441
Residuals	13102.1	115		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Step 2: Fit ANCOVA Model

```
# Fit the ANCOVA model (without interaction)
ancova_model <- lm(LONGEV ~ THORAX + treatment, data = partridge)

# Get more detailed summary
summary(ancova_model)
```

Call:

```
lm(formula = LONGEV ~ THORAX + treatment, data = partridge)
```

Residuals:

Min	1Q	Median	3Q	Max
-26.189	-6.599	-0.989	6.408	30.244

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-46.055	10.239	-4.498	1.61e-05
THORAX	135.819	12.439	10.919	< 2e-16
treatmentOne virgin female daily	-3.929	2.997	-1.311	0.192347
treatmentEight virgin females daily	-1.276	2.983	-0.428	0.669517
treatmentOne inseminated female daily	-10.946	2.999	-3.650	0.000391
treatmentEight inseminated females daily	-23.879	2.973	-8.031	7.83e-13

(Intercept) ***
THORAX ***
treatmentOne virgin female daily
treatmentEight virgin females daily
treatmentOne inseminated female daily ***
treatmentEight inseminated females daily ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.51 on 119 degrees of freedom

Multiple R-squared: 0.6564, Adjusted R-squared: 0.6419

F-statistic: 45.46 on 5 and 119 DF, p-value: < 2.2e-16

```
# View ANOVA table
anova(ancova_model)
```

Analysis of Variance Table

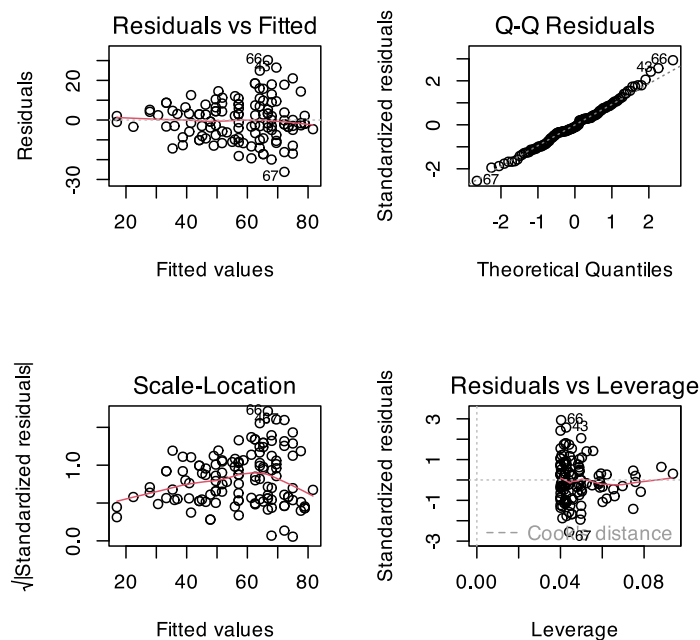
Response: LONGEV

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
THORAX	1	15496.6	15496.6	140.293	< 2.2e-16 ***
treatment	4	9611.5	2402.9	21.753	1.719e-13 ***
Residuals	119	13144.7	110.5		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Step 3: Check Assumptions

```
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(ancova_model)
```



Step 4: Calculate Adjusted Means

```
# Get adjusted means using emmeans
adjusted_means <- emmeans(ancova_model, "treatment")
adjusted_means
```

treatment	emmean	SE	df	lower.CL	upper.CL
No females	65.4	2.11	119	61.3	69.6
One virgin female daily	61.5	2.11	119	57.3	65.7
Eight virgin females daily	64.2	2.10	119	60.0	68.3
One inseminated female daily	54.5	2.11	119	50.3	58.7
Eight inseminated females daily	41.6	2.12	119	37.4	45.8

Confidence level used: 0.95

Step 5: Pairwise Comparisons

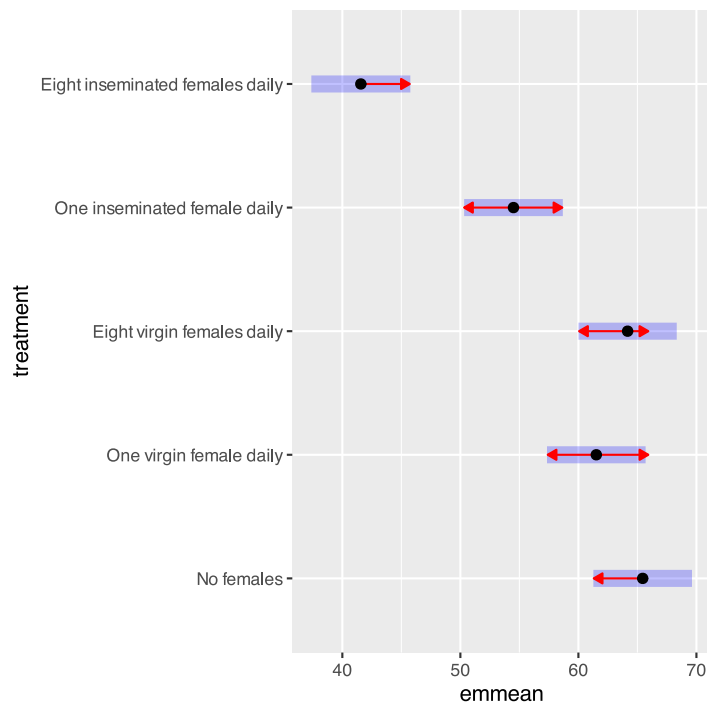
```
# Pairwise comparisons of adjusted means  
pairs(adjusted_means, adjust = "tukey")
```

contrast	estimate	SE
No females - One virgin female daily	3.93	3.00
No females - Eight virgin females daily	1.28	2.98
No females - One inseminated female daily	10.95	3.00
No females - Eight inseminated females daily	23.88	2.97
One virgin female daily - Eight virgin females daily	-2.65	2.98
One virgin female daily - One inseminated female daily	7.02	2.97
One virgin female daily - Eight inseminated females daily	19.95	3.01
Eight virgin females daily - One inseminated female daily	9.67	2.98
Eight virgin females daily - Eight inseminated females daily	22.60	2.99
One inseminated female daily - Eight inseminated females daily	12.93	3.01

df	t.ratio	p.value
119	1.311	0.6849
119	0.428	0.9929
119	3.650	0.0035
119	8.031	<.0001
119	-0.891	0.8996
119	2.361	0.1336
119	6.636	<.0001
119	3.249	0.0129
119	7.560	<.0001
119	4.298	0.0003

P value adjustment: tukey method for comparing a family of 5 estimates

```
# Plot adjusted means with confidence intervals  
plot(adjusted_means, comparisons = TRUE)
```



Part 3: Example with Heterogeneous Slopes

Let's look at an example where slopes are NOT homogeneous using sea urchin data.

```
# Create simulated sea urchin data with heterogeneous slopes
set.seed(345)
n <- 72 # 24 urchins per group

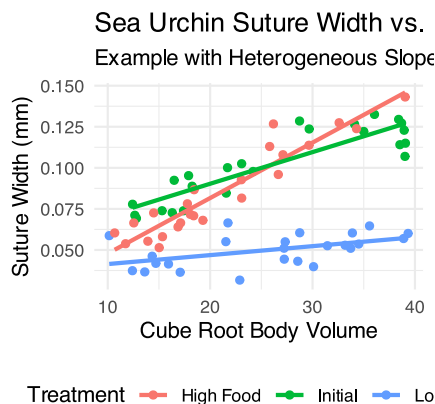
# Create data frame
treatments <- rep(c("Initial", "Low Food", "High Food"), each = n/3)
volume <- c(
  runif(n/3, 10, 40), # Initial
  runif(n/3, 10, 40), # Low Food
  runif(n/3, 10, 40)  # High Food
)

# Create suture width with different slopes for each treatment
suture_width <- ifelse(
  treatments == "Initial", 0.05 + 0.002 * volume,
  ifelse(
    treatments == "Low Food", 0.04 + 0.0005 * volume,
    0.02 + 0.003 * volume # High Food
  )
) + rnorm(n, 0, 0.01)

urchin_data <- data.frame(treatment = treatments, volume = volume, suture_width =
suture_width)

# Plot the data with regression lines
ggplot(urchin_data, aes(x = volume, y = suture_width, color = treatment)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Sea Urchin Suture Width vs. Volume",
       subtitle = "Example with Heterogeneous Slopes",
       x = "Cube Root Body Volume",
       y = "Suture Width (mm)",
       color = "Treatment") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

`geom\_smooth()` using formula = 'y ~ x'



Test for Homogeneity of Slopes

```
# Fit model with interaction
urchin_model <- lm(suture_width ~ volume * treatment, data = urchin_data)
Anova(urchin_model, type = 3)
```

Anova Table (Type III tests)

Response: suture_width

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	0.0005253	1	5.91	0.01778 *
volume	0.0151663	1	170.64	< 2.2e-16 ***
treatment	0.0020070	2	11.29	6.064e-05 ***
volume:treatment	0.0062129	2	34.95	4.453e-11 ***
Residuals	0.0058662	66		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Result: With $p < 0.05$, we have heterogeneous slopes! Standard ANCOVA would be inappropriate here.

What to do with Heterogeneous Slopes

When slopes are not homogeneous, you have several options:

```
# Option: Analyze groups separately
initial_model <- lm(suture_width ~ volume, data = filter(urchin_data, treatment ==
"Initial"))
low_food_model <- lm(suture_width ~ volume, data = filter(urchin_data, treatment == "Low
Food"))
high_food_model <- lm(suture_width ~ volume, data = filter(urchin_data, treatment == "High
Food"))

# Summary for each group
initial_model
```

Call:

```
lm(formula = suture_width ~ volume, data = filter(urchin_data,
treatment == "Initial"))
```

Coefficients:

(Intercept)	volume
0.051785	0.001926

low_food_model

Call:

```
lm(formula = suture_width ~ volume, data = filter(urchin_data,
treatment == "Low Food"))
```

Coefficients:

(Intercept)	volume
0.0359532	0.0005453

```
high_food_model
```

```
Call:
lm(formula = suture_width ~ volume, data = filter(urchin_data,
  treatment == "High Food"))
```

```
Coefficients:
(Intercept)      volume
  0.014077      0.003376
```

Summary Checklist for ANCOVA

When conducting ANCOVA, always follow these steps:

💡 ANCOVA Checklist

1. **Visualize your data** - plot response vs covariate, colored by groups
2. **Test homogeneity of slopes** - fit model with interaction term
 - If $p > 0.05$: proceed with ANCOVA
 - If $p < 0.05$: use alternative approaches
3. **Fit ANCOVA model** - response ~ covariate + factor
4. **Check assumptions** - use diagnostic plots
5. **Interpret results** - focus on adjusted means, not raw means
6. **Conduct post-hoc tests** - pairwise comparisons if needed
7. **Visualize results** - show adjusted means with confidence intervals

Key Points to Remember

- **ANCOVA increases power** by accounting for covariate variation
- **Adjusted means** are what we compare, not raw group means
- **Homogeneity of slopes** is the most critical assumption
- **Parallel lines** in your plot suggest homogeneous slopes
- **Non-parallel lines** indicate heterogeneous slopes - use alternative methods

! Key Points from ANCOVA Analysis

1. **Test homogeneity of slopes first** - this is the most critical assumption
2. **ANCOVA compares adjusted means** at the mean value of the covariate
3. **Increases statistical power** by removing variation due to the covariate
4. **Choose appropriate methods** based on whether slopes are homogeneous
5. **Visualize your results** clearly showing the relationship between variables
6. **Check all assumptions** using diagnostic plots
7. **Interpret in biological context** - what do the adjusted means tell us?

Remember: The covariate should be measured independently of the treatment and should not be affected by the treatment itself!